A Galaxy Law Tied to Dark Energy, and a Map of What It Leaves Unexplained

Working note (PAPER67), version 1.0, 2026-11-02. AI-assisted research programme; not peer reviewed. $\kappa = \frac{1}{2}$ is fitted. The theory is not closed. Nothing here says the data favour this framework over any other.

Abstract

We report what is left of a research programme after its own final grade. A galaxy law, $g_N = g^2/(g + a_0)$ with $a_0 = \kappa c\sqrt{G\rho_{\rm DE}}$ and $\kappa = \frac{1}{2}$ fitted, describes isolated disc galaxies well on the canonical footing: SPARC rms 0.0541 dex at median Υ 0.52, no tilt with speed, no slope with surface brightness at fixed speed. The law is a restatement of MOND's simple form with a proposed origin for its scale. Earlier authors proposed that origin, and the scale's tie to dark energy, as distinct from cH_0, has no decisive test yet.

The law fails IC 1613, several Local Group classes, cluster shear and X-COP, whose hydrostatic masses it misses by a factor of 2.93, and the unscreened Solar System. Its Milky Way result depends on a vertical form it doesn't supply, and at 60–150 km s^{-1} it asks for about twice the stellar mass estimated from photometry. It has no action and supplies no dust-like term for the CMB. To cover those we developed a cold dark energy sector. On measured data every graded candidate fails, including the law alone, and the sector is now retired. We publish the scorecard, the two walls that stopped it, and twelve requirements any successor must meet. This is numerical evidence for the families tried, not a no-go theorem. We close with dated tests that can still kill the law.

1 Scope

This is a programme report, not a complete theory. Three owner decisions on the record set its form: the a_0 floor was removed from the cold sector (2026-10-31), the search for a derived threshold stopped (2026-11-01), and the cold sector was retired (2026-11-01). Earlier deposits stand as dated records. The two footings, canonical $a_0 = 9.3603 \times 10^{-11}$ and alternative 1.1312×10^{-10} m s^{-2}, are never pooled. This paper's audit script checks the quoted numbers against committed files. It doesn't check the prose.

2 The law, and whose idea it is

$$g_N = \frac{g^2}{g + a_0}, \qquad a_0(z) = \kappa c\sqrt{G\rho_{\rm DE}(z)}, \qquad \kappa = \tfrac{1}{2} \text{ (fitted)}.$$

Credit. The first relation is Milgrom's modified dynamics in its simple form [1]. The numerical nearness of a_0 to cH_0 and to $c^2\sqrt{\Lambda}$ is his observation [1, 2]. Limbach, Psaltis & Özel [3] proposed the tie to the dark energy density. None of this originated here.

κ isn't derived. Of 537 attempts on the record, none derives $\kappa = \frac{1}{2}$. It isn't singled out by SPARC either. A profile likelihood over 175 galaxies has a free best fit of 1.0766×10^{-10}. Against it, $\kappa = \frac{1}{2}$ on $\rho_{\rm DE}$ reads $\Delta\chi^2$ 63.90 (1.82σ with galaxy-clustered errors), the alternative footing 7.04 (0.60σ), and Milgrom's own $cH_0/2\pi$ 5.34 (0.53σ). So Milgrom's value fits at least as well as ours.

What this programme adds: a reconstruction of $a_0(z)$ from raw BAO and supernovae with a stated ruler, the graded comparison below, and the failure record.

The band, and what it rests on. The reconstruction gives $a_0(z)/a_0(0)$ of 0.91 [0.76, 1.03] at $z = 1$, 0.94 [0.71, 1.12] at $z = 1.5$ and 0.91 [0.33, 1.21] at $z = 2.33$ (68 %), against 1.76, 2.35 and 3.49 for the rival $a_0 \propto H(z)$. Constant dark energy isn't rejected (1.97σ, primary). **This band isn't free of the retired sector's input:** it takes $\rho_m(z) = \omega_m(1 + z)^3$ with a cold mass from the CMB, and with positivity alone, no CMB and no H_0, its check fails. The law doesn't explain that matter term (Section 5).

The distinctive claim is untested. At $z \approx 0$ the law can't be told from $a_0 \propto cH_0$. Separating them at 3σ needs a_0 to 0.055 dex at $z = 1$ and 0.084 dex at $z = 2.33$. The best public sample reaches 0.4–0.5 dex.

Table 1: The law on isolated systems. FG1 is the frozen final grade; a clean-room regrade (CRG1) reproduced its table, 289 match, 6 close, 0 disagree, mainly a test of code.

line	canonical	alternative	lane
SPARC rms, median Υ (line 0.50–0.80)	0.0541, 0.52	0.0537, 0.44 **fails the line**	FG1
tilt with speed	+0.02	−0.02	FG1
LITTLE THINGS, median E	−0.025	−0.064	FG1
63 groups, Z	+0.31	−0.31	FG1
slope with surface brightness, 60–120 km s^{-1}	-0.016 ± 0.061, flat	-0.023 ± 0.063, flat	HS1
local a_0, MeerKAT against SPARC	+0.032 dex (line 0.15)		PAPER60

3 What it does in isolated galaxies

A caution on the Υ line. The 0.50–0.80 line was first written for this law, and its lower edge is a population model's central value, not a measurement. So passing it is weaker than it looks; Section 4 sets it against measured masses.

Surface brightness is the strongest result. At fixed speed, a halo set by mass alone can't follow the disc: every version of the retired sector's halo tilts, −0.17 to −0.29 per dex, and the law doesn't. That's a real discriminator on the data used, between the law and these halos only.

Isolated-galaxy lensing is shape, not a clean amplitude. KiDS-1000 prefers the law's shape over the retired halos ($\Delta\chi^2$ +366.0 against the best native halo). But the law's amplitude depends on colour and on the baryon inventory: blue lenses read $A = 1.016 \pm 0.086$ in one lane and 0.878 in a later forward model that adds a hot gas inventory, and red lenses 1.587 and 1.313. No tested model fits these profiles in absolute terms.

4 Where the law fails

- **Milky Way.** The law gives only the in-plane force, so the vertical test needs a form it doesn't supply. The field-equation form (AQUAL) is INFEASIBLE, 0 cells, on both footings. The round rule, which has no field equation behind it, passes in 1 in-prior cell (alt 6) (LW1). On an older, weaker gate, 7 of 175 cells pass (alt 4), all with a short disc and the lightest local column; those have $K_z(R_0)$ 76.7–80.9 against 66.4 ± 2.2, +4.7 to $+6.6\sigma$, and leave the outer slope excluded (Z −3.88). FG1 counts this as FAIL.
- **IC 1613.** $E = -0.593$, $Z = -7.9$; it fails under every halo-free inclination. One galaxy: disfavoured, not excluded.
- **Local Group classes.** Z: MW ultra-faints +5.22, bright M31 +2.20, other M31 +2.29, MW classicals +1.69, field +0.57. Adding the external field makes it worse: ultra-faints +6.76, bright M31 +3.97. And at fixed M_V the data are flat in host field, -0.05 ± 0.19 dex per dex, where AQUAL with its external field predicts −0.76, so neither version describes these satellites (LW1).
- **Measured stellar masses, 60–150 km s^{-1}.** The law's Υ is 0.62 and 0.65 where GSWLC stellar masses give 0.277 and 0.304 (Z +3.4, +5.6). These are estimators against W1 luminosity, not a 3.6 μm measurement. Gas dominance doesn't excuse the 100–150 bin.
- **Disc weighings.** On this law's own form, DiskMass σ_z reads round rule $r = 1.239$ (alt 1.204), outside 0.90–1.10, a FAIL as frozen (LW1). Earlier lanes read $r = 1.250$ and 1.342 with a different interpolating function. Same-population dispersions [7, 8, 9] correct r by c of 1.414–1.517, giving 0.824–0.884 one-zone and, with exact Jeans and measured thickness, inside the line. But the sources' own tables run c from about 1.24 at the relevant radius to 1.88 in integrated light, so the question is open (OD1, NOT SEPARATED). NGC 628 reads $\Upsilon_{3.6} = 0.75 \pm 0.18$ and NGC 6946 about 0.4.
- **The Solar System.** The unscreened law fails Cassini's Q_2 by 2.44–4.59 times the 2σ line (alt 2.85–5.19). The pass in earlier reports belongs to a screened covariant theory. That line's final verdict (v3s) was that it holds as a galaxy law and fails as gravity plus cosmology. So the law as written here needs a screen it doesn't contain.

5 What it leaves unexplained

- **Cluster cores and shear.** From measured gas, in 318 eROSITA clusters with raw KiDS shear, the law is short by $A = 1.91 \pm 0.12$ at 0.5–1.2 Mpc (alt 1.75 ± 0.11) and matches at 1.2–3 Mpc, $A = 1.01 \pm 0.09$. On the measured chain it fails 10 of 15 keys on both footings, G-OUT-M3 +4.76, joint p 3e-8. On this statistic that is worse than the retired native profile (LW1).
- **X-COP.** The law's own number is +0.467, rms 0.476: $M_{\rm HSE}/M_{\rm law}$ of 2.93 (alt 2.72), all 7 clusters short. Taking 0.12 dex off the hydrostatic masses still leaves 2.23 (alt 2.06). An earlier scorecard showed another rule's row here; LW1 corrects it.
- **What is missing in clusters, as a description.** Inside about R_{500} the deficit is near 8 times the gas at 0.2–0.5 R_{500} and 5 times at 0.5–1 R_{500}, with no detected mass dependence, no BCG link and no preferred scale found, picked from 31 looks (LM1). Trained on KiDS, it predicts X-COP's hydrostatic masses out of sample, but HSC lensing reads it about 1.7 times higher, and tracking gas can't be told from tracking M_{500}, so its frozen label is NOT (GM1, which issued labels after a failed check and says so). No mechanism follows.
- **Outer shear against gas mass.** $\Delta\Sigma_{\rm OUTER} \propto M_{\rm gas}^{s}$ with $s = 0.90 \pm 0.21$, against the law's 0.47 and Newton's 0.84. The amplitude agreement above is an average over a slope the law misses.
- **The CMB and growth.** a_0 is absent from the linear FRW equations, so the law can't act there. The record has no CMB pass without a dust-like component. With one, a window exists at $z_{\rm on} \in [3 \times 10^4, 10^6]$ on the channels run. Retiring the sector leaves this unanswered. **The law offers no replacement, and we don't claim it needs none.**
- **The matter term.** The law takes $\rho_{\rm DE}$ from an expansion history whose matter term is several times the baryons. That term is unexplained here.
- **An action.** None is selected, and the a_0 sector's action stays open. Attempts on record are this programme's own versions on AeST [4] and Khronon [5], and don't test those theories as published. v9, an AeST embedding, fails the preferred-frame bounds, $\alpha_1 = -2(K_B + 2)$. v3, Khronon with an added term, fails linear growth: the published cap $\mu_K \leq 14.8\,h$/Mpc sits 24–172 times below what σ_8 needs. v4, AeST with that term, is ill-posed where the aether twists and runs away on FRW for the dust shape tried. v3s, v3 with switched stiffness, is frozen as a galaxy law that fails cluster cores, linear growth, hosted dwarfs and the Milky Way disc.

6 The cold sector: tried and retired

What it was. A gravitating component with, at linear order, $w = 0$, $c_s = 0$, geodesic motion and an abundance set by an integration constant. A referee will read those as the linear behaviour of cold dark matter, and that's correct; it was never proposed as a particle, and the programme's no-particle rule is unchanged. Its candidates: C1, the native collapse rule N; C2, N with the measured gravitational response; C3, a basin split at $T \approx 10^{12.3}\,M_\odot$; C5, the native shape with the measured response; with C4, the law alone, as the control.

Table 2: FG1, canonical, with C4's X and DM from LW1 and C5 from HM1. P pass, F fail, – not graded, which isn't a pass. No winner is declared, because the lines aren't independent.

	S	Tl	LT	G	K	X	Ln	Dw	MW	DM	IC	alt
C1 native N	P	F	F	P	F	P	F	F	F	F	F	no a_0, same
C2 N + response	F	F	F	P	F	F	F	–	F	P	–	no a_0, same
C3 basin split	P	P	P	P	F	P	P	P	F	–	F	also S, Dw fail
C4 law alone	P	P	P	P	F	F	P	F	F	F	F	also S fails
C5 native + response	F	F	F	F	F	F	F	F	P	P	F	not listed

S SPARC, Tl tilt, LT LITTLE THINGS, G groups, K 15 cluster keys, X X-COP, Ln KiDS isolated, Dw Local Group, MW Milky Way, DM DiskMass, IC IC 1613. C4's MW cell depends on the vertical form. C5's row is a reading of its numbers: HM1 printed NO VERDICT as frozen because one mass check failed, and an amendment moves no key by over 0.001. FG1's tag [GAS-CONDITIONAL]: at X-COP's gas scale +0.111 the native rule reaches 15 of 15 keys, rescuing K only; the measured scale is $+0.005 \pm 0.017$.

Wall 1: spirals at 150–250 km s^{-1}. Whatever is dust in the CMB falls into galaxies, and a live disc measures it contracting: $f_Q = 0.872$ [0.54, 1.07] (LC1), 0.946 [0.78, 1.08] with the disc grown from turnaround (PZ1). The SPARC side needs net f of about 0–0.2 at the 0.50 Υ line (RM2, six of 75 cells, three fitted numbers). This is the

known Tully–Fisher zero-point problem [6] in measured numbers. A cored halo's response is somewhat weaker, $f_B = 0.753$ [0.729, 0.777] (DR8), not enough. Whether the 0.50 line is itself too high is open (Section 4, UL1, OD1); moving it wouldn't clear tilt or LITTLE THINGS.

Wall 2: outer cluster shear on measured gas. G-OUT-M3 reads 1.56 against 1.04, $Z = +4.41$ statistical, about +3.03 to +3.51 with systematics (M3A1). It's not an artefact on anything measurable from disk, nor the model's edge, and native two-halo supplies 3–11 % of +21 to +27 %.

No derived switch. The basin split's T was located, never derived. Three frozen attempts failed: 13.94 for the turned-around mass and 13.54–13.89 for a clock, against a window of 12.19–12.39 in $\log M$, and a candidate action had no link to mass. In shift-symmetric $P(X)$, a turn-on has $c_s^2 < 0$ at all 14 grid points, a result within that family (BOLTZ1).

Table 3: Doors tried in the closing push, by each lane's own label. Full list: the closing ledger.

idea	lane	result
density term in the law	DR1	SPARC BREAKS, CLUSTERS NOT REACHED
phase-space cap on response	DR7	BITES, FAILS [NOISE-LIMITED]
retention replaces the mass line	RET1	DERIVED LINE FAILS, FITTED ϵ FAILS
barotropic halo	HB1	MOVES, FAILS
fresh hunt, 19 routes	MH1	no mechanism survives whole
starting point	SP3	no response-on start reaches Υ's line in the owner's grid
symbolic search	SL1	TRAIN ONLY, 0 survivors

Gas retention: the mass method decides it, and it isn't settled. A gas-fraction line from hydrostatic masses, fitted without X-COP or shear, takes the native profile from 12 to 14 of 15 keys with X-COP passing, G-OUT-M3 +3.48, +3.09, with frozen label NOT RETENTION by one cell (FR1). Okabe's HSC masses on eRASS1 gas read 0.216 dex below that line, -4.4σ; through the same grader that overshoots, 4 of 15 keys, G-OUT-M3 −2.54, X-COP failing. A second HSC reading, Umetsu on XXL gas, sits on the hydrostatic line instead (FR3). Neither line passes all 15 keys and X-COP. Both lanes were frozen after the retirement, weren't blind and don't touch galaxies.

Status. Retired by the owner on 2026-11-01. One lane frozen before that decision, a per-galaxy action response (AR1), had not finished when this version was deposited, and its result isn't here. No later result can reinstate the sector; only the owner's decision can. What survives from this work as fact, not as a theory: native collapse passes X-COP on hydrostatic masses, +0.013, rms 0.075, and one adiabatic response has the opposite sign in discs and cluster cores.

7 What a successor must do

The collisions. Rows 4 and 11: dust in the CMB falls into galaxies. Rows 1–3 and 10: the amount needed runs from about 0.8 of native on SPARC to 1.22–1.80 on clusters. Rows 7 and 8: systems of the same mass want different things. Selectors closed on inputs tested: potential, σ, acceleration, environment, formation epoch, gas state, bars. None of this is a theorem.

One family left outside. A 5–11 eV fermion would address rows 4 and 11 directly. Its forest kill isn't on the current footing: a 1-D proxy against a ΛCDM yardstick falls 1000–5000 times short, undecided on raw forest data. Other liabilities stand, and it wouldn't fix rows 6 or 7. It's outside the no-particle rule.

8 Dated tests, and their real power

- **ALMA U4_28473, public 2026-11-06 02:42:31 UTC** (PAPER66 v1.2, DOI 10.5281/zenodo.23868979). It tests this galaxy's mass budget, not $a_0(z)$, and one disc excludes no model.
- **GS4_24110,** [CI](2–1), 2026-11-28 by earlier metadata, to be rechecked. Every $n = 1$ against $n = 2$ pair is at or below 0.51σ, so it can't separate them.

Table 4: Requirements from combining the failures. Older rows are on older chains, and FG1 supersedes any it covers. Coverage: an inventory that opened about 90 of about 330 lanes.

#	must	law	native halos
1	keep galaxies tight	**0.0541**	0.0682 / 0.0694
2	stay flat with surface brightness	−0.016	−0.173
3	sit on Υ at every speed	0.52, alt 0.44	0.75, running 1.20 to 0.57
4	show little net contraction in spirals	not needed	needs $f \leq 0.2$, measured 0.872
5	fit dwarfs	−0.025	+0.138; 0 of 5 classes
6	fit IC 1613	−0.593	−0.474
7	fit the Milky Way	depends on vertical form	asks 4e10 $M_\odot$ inside R_0
8	fit isolated lenses	shape preferred	+366.0
9	add mass in cluster cores	A 1.91	G-IN −0.49, G-MID +2.17
10	match outer cluster shear	G-OUT-M3 +4.76; slope 0.47 against 0.90	G-OUT-M3 +4.41
11	act as dust in the CMB	no cold term	$z_{\rm on} \in [3 \times 10^4, 10^6]$
12	have an action	open	$P(X)$ dust limit only

- **Gaia DR4, about December 2026** (pre-registered 2026-07-16, with later amendments). Wide binaries: Newton 1.000, an ungated MI band 1.05–1.10, AQUAL point 1.137, MOND's benchmark 1.33; the DR3 dry run's 1.205 ± 0.035 is evidence for nothing. An amendment registers a gated branch for which 1.000 inside 30 kAU is also predicted, so that branch needs its own test beyond 50 kAU, about 2000 clean pairs per bin. The solar-acceleration dipole's kill bands are void under the amended theory, so that front returns no verdict. *Reading, not a frozen line:* these bands were written for the action-carrying framework. The bare law here contains no gate, so a Newtonian result at 2–30 kAU would count against it.
- **SP1, SP2 sealed lensing predictions.** Scored as frozen, including the retired route's rows.
- **Named, not scheduled:** one clean deep-MOND a_0 point at $z \approx 2.5$ to about 0.10 dex. That's the measurement that would actually test the dark energy tie.

9 Method

Each lane commits frozen criteria alone before any script or number, runs a MUTATE twin that must fail, reruns byte-identically and publishes failures under frozen labels. Most lanes were not blind. Grading was done by the same programme that proposed the models; the clean-room regrade tested code, not independence of definitions. No outside group has reproduced these results.

Not claimed

A complete, closed or unified theory. That the data favour this framework. κ, T or any abundance as derived. That the tie of a_0 to dark energy originated here. That the requirements list is a no-go theorem.

Data and code

All lanes named here are in `rebuild_2026_10_22/` and `campaign_fresh_gravity/`. Scorecard: `FG1_final_grade`. Law's scorecard: `LW1_law_scorecard`. Ledger, still a working draft: `CLOSING_LEDGER_2026-11-01_DRAFT.md`. Audit: `PAPER67_audit.py`.

References

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